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[22CSBS120]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

SOFTWARE ENGINEERING

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

1.	**a.**	List out the applications of Software Engineering.	L1	CO1	**02**
	b.	What is software planning?	L1	CO1	**02**
	c.	Define Capability Maturity Models.	L1	CO1	**02**
	d.	Write the concept of encapsulation.	L1	CO1	**02**
	e.	What is black box testing?	L1	CO1	**02**
		PART-B		**6 X 5=30**	
2.	**a.**	Discuss about the importance of software quality and its timely availability for the successful completion of the large software projects.	L2	CO2	**05**
		OR			
	b.	Discuss in detail that how software engineering evolved as a discipline.	L2	CO2	**05**
3.	**a.**	What is feasibility study? Discuss at least two types of feasibility in brief.	L2	CO2	**05**
		OR			
	b.	List out the agile methods. Discuss in brief.	L2	CO2	**05**
4.	**a.**	What is software Quality Dilemma? Discuss its various aspects in details.	L2	CO2	**05**
		OR			
	b.	Explain Software Reliability and list out the software reliability models.	L2	CO2	**05**
5.	**a.**	What is scrum? Discuss the role of scrum.	L3	CO3	**05**
		OR			
	b.	What do you mean by extreme programming? Discuss in brief.	L2	CO2	**05**
6.	**a.**	Explain Class Responsibility Collaborator model? How this is used?	L3	CO3	**05**
		OR			
	b.	What is Software Requirement Specification? Explain in details.	L2	CO3	**05**
7.	**a.**	What do you mean by requirement documentation with use case?	L2	CO2	**05**

		OR			
	b.	What is unit testing? Discuss in brief.	L2	CO2	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Explain Software Development Life Cycle in brief. Elaborate on the various issues of SDLC.	L1	CO1	**10**
		OR			
	b.	What do you mean by programming in small and programming in large? Differentiate between these two with an example.	L2	CO1	**10**
9.	**a.**	Discuss in details about the measurement of software size.	L2	CO2	**10**
		OR			
	b.	What is agile software engineering? Discuss in detail about the concepts of agile methods.	L3	CO2	**10**
10.	**a.**	How do we measure the quality of the software? Discuss in brief about ISO 9126 quality factor.	L2	CO2	**10**
		OR			
	b.	Explain in detail about the software engineering historical development.	L1	CO1	**10**
11.	**a.**	What do you mean by software metrics model? How to track and measure those metrics? Explain with an example in detail.	L2	CO2	**10**
		OR			
	b.	With suitable illustration Discuss about the concept of abstraction and information hiding.	L2	CO2	**10**
12.	**a.**	Explain the various types of system testing.	L1	CO1	**10**
		OR			
	b.	Explain black box and white box testing and discuss their outcomes.	L2	CO2	**10**
13.	**a.**	Discuss in brief about the concept of verification and validation.	L3	CO2	**10**
		OR			
	b.	Write a note on **i.** Integration testing **ii.** Acceptance testing	L3	CO3	**10**